



UNIVERSITY OF VOCATIONAL TECHNOLOGY
IT304082 – Database Implementation
Final Assignment Brief

Assignment Information

Field	Details
Module	Database Implementation
Module Code	IT304082
Team Size	Maximum 5 members per group
Duration	2 Weeks
Technology	PHP and MySQL ONLY
Submission	GitHub Repository Link + PDF Document

1. Overview

Build a simple website using PHP and MySQL on a topic of your choice. The main goal of this assignment is to demonstrate your MySQL database knowledge. The website is simply the interface to show your database working in a real application.

You are free to choose any application domain, for example:

- Library Management System
- Student Result System
- Hotel Booking System
- Online Shop
- Hospital Appointment System
- Any other topic with lecturer approval

Note: Choose something you understand well. The simpler the website, the more time you have to focus on the database.

2. 3-Tier Architecture

Your project must follow the 3-Tier Architecture model. Each tier must be clearly separated in your project folder structure and explained in your documentation.

Tier	Description	Technology
Tier 1	Presentation Layer — Web pages the user sees	HTML, CSS, JavaScript
Tier 2	Application Layer — PHP logic connecting UI to database	PHP (core only, no frameworks)
Tier 3	Data Layer — All data stored and managed here	MySQL

3. What to Build — Website (Keep it Simple)

Your website must include:

- A login page with at least two roles (e.g., Admin and User)
- Pages to Add, View, Update, and Delete data (CRUD) for your main entities
- One report or summary page that displays meaningful data from the database (totals, counts, lists)
- Clean and readable UI — it does not need to be fancy, just working and well organized

4. What to Build — MySQL (Main Focus)

This is where most of your marks are. Your database must demonstrate the following:

MySQL Feature	Minimum Requirement
Tables	At least 5 related tables with proper relationships
ER Diagram	Full diagram showing all tables and relationships
Normalization	Minimum 3NF — must be explained in your document
Keys & Constraints	All tables must have proper primary keys, foreign keys, and constraints
Stored Procedures	At least 3 — must be called and used in your PHP application
Triggers	At least 2 — must fire automatically on INSERT / UPDATE / DELETE
Views	At least 2 — used for report or summary pages
Indexes	Applied on columns that are frequently searched or filtered

Transactions	Used in at least 1 place where multiple tables are updated together
SQL Queries	At least 8 meaningful queries using JOINS, GROUP BY, subqueries, aggregations

5. PHP Requirements

The website just needs to work. Keep it straightforward:

- Connect to MySQL using PDO or MySQLi with prepared statements
- Basic session-based login and logout
- Server-side validation on all forms (check empty fields, correct data types)
- Your PHP must actually call and use your stored procedures, views, and triggers — they cannot just sit in the database unused

6. Limitations and Rules

The following are strict rules. Violating them will result in mark deductions:

Rule	Detail
PHP frameworks	Not allowed — use core PHP only (no Laravel, CodeIgniter, etc.)
Raw SQL in PHP	Not allowed — use prepared statements (PDO or MySQLi)
Unused database features	Triggers, procedures, and views must be actively used in the website
Copy-paste code	You must be able to explain every part during Q&A
Late GitHub commits	Must show regular commits — single last-minute push will be penalized

7. GitHub Repository Requirements

Your GitHub repository must be public and contain the following:

- Complete source code of the website
- All SQL scripts — schema.sql, data.sql, and a file for procedures, triggers, and views
- Screenshots of all main pages inside a /docs folder
- A README.md file with the project description, setup instructions, list of all functionalities, and the team contribution table

Contribution Table — must be included in your README.md:

Member Name	Student ID	What They Did
Member 1	XXXXXXX	e.g. Database design, stored procedures, triggers
Member 2	XXXXXXX	e.g. Database design, stored procedures, triggers
Member 3	XXXXXXX	e.g. Database design, stored procedures, triggers
Member 4	XXXXXXX	e.g. Database design, stored procedures, triggers

Important: GitHub commit history will be reviewed. Members with little or no commits may lose individual marks.

8. PDF Document Requirements

Submit a short PDF document (maximum 20 pages) containing the following sections:

- Cover page — team name, student IDs, and application topic
- Introduction — what your website does
- ER Diagram
- Normalization explanation — show the process from 1NF to 2NF to 3NF
- List of all MySQL features implemented — with a brief explanation of each procedure, trigger, and view
- Screenshots of all main pages
- Team contribution — who did what, clearly stated for each member
- Challenges faced and how you solved them

9. Marking Criteria

Component	Marks	Weight
Database Design — ER diagram, normalization, schema, relationships, constraints	20	20%
MySQL Features — Stored procedures, triggers, views, indexes, transactions, query complexity	40	40%
PHP & Working Website — CRUD works, login works, database features used in the site	20	20%
Documentation & GitHub — README, PDF, commit history, contribution table	10	10%
Q&A — Every member must explain their own contribution	10	10%
TOTAL	100	100%

10. Submission

Submit the following via the course portal:

- GitHub repository link (must be public)
- PDF document

Submission Deadline: 2026 May 3

All team members must be present for the live demo and Q&A session.

For questions, contact your module lecturer during consultation hours.